Bayesian Knowledge-Driven Ontologies for Synthetic Data

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Testing whether medical knowledge can guide synthetic patient-data generation to fill coverage gaps and improve heart-disease prediction for underrepresented groups.

1. The Problem and Organizational Context

Clinical machine learning systems fail at long-tail populations. A disease-detection model receives thousands of records for common presentations and almost none for rare subtypes or underserved subgroups, resulting in costly missed diagnoses. Existing paths to more data are slow or lossy: data-use agreements take weeks, HIPAA Safe Harbor strips the quasi-identifiers that make rare cases findable, and SMOTE only interpolates between existing examples.

Commercial synthetic-data platforms (e.g., Gretel, MDClone, MOSTLY AI) optimize for statistical fidelity to the supplied distribution. When that distribution is incomplete, fidelity reproduces the very gaps that cause failure.

This proposal reframes the platform around **knowledge** instead, using **Bayesian Knowledge-driven Ontologies (BKOs)**, developed at Thayer by Santos, Jurmain, and Ragazzi (PLOS ONE, 2024). BKOs integrate description logic with Bayesian Knowledge Bases; unlike Bayesian networks. BKOs treat incompleteness as a property of the representation.

2. Intended User

The primary users are machine learning engineers or clinical data scientists at a digital health startup or an academic laboratory, with clinicians as secondary users.

As an initial evaluation setting, we predict the presence of heart disease on the UCI Heart Disease dataset, operationalizing underrepresentation as fewer than 20 training records per age–sex subgroup-label combination. This threshold is a prototype convenience criterion and does not claim to define clinical rarity. A complementary Synthea experiment will assess gap detection by deliberately ablating selected cases from the generated population.

3. Proposed AI Approach

Represent. The objective is to load MONDO and HPO as BKO fragments and add the user's dataset as a second fragment weighted by observed frequency. Ontology concepts will be mapped to dataset fields, starting with the disease label and chest-pain type, and each mapping's source, encoding, and limits will be documented, with clinician review where available. Numeric fields will be governed by separately specified range constraints, and concepts without a defensible corresponding field will be excluded from scope.

Detect. The BKO instantiation algorithm will be applied to identify statements implied by the ontology but unsupported by the data. Because Santos et al. prove these ungrounded statements contribute nothing to the model's distribution, they will serve as a formal definition of a dataset gap instead of a heuristic. Generation will be guided by the relevant ontology relationships, and each candidate record will be validated before augmentation. Count-based and BKO-detected gaps will be evaluated separately.

4. Tools and Architecture

The front end will be built in Streamlit over a Python backend.Ontologies will be accessed through owlready2, pronto, and OLS4, and records will be generated through the Anthropic Messages API. CTGAN (Conditional Tabular GAN), TVAE (Tabular Variational AutoEncoder), SMOTE (Synthetic Minority Over-sampling Technique), or ADASYN (Adaptive Synthetic Sampling) will be used as baselines, and evaluation will be performed with scikit-learn, SDMetrics, and synthcity. Code will be managed in a GitHub repository. Both datasets (UCI Heart Disease and Synthea) are openly available. The principal risk is that no public BKB library exists, so the inference layer will be built by the team.

5. Testing, Validation, and Success Criteria

Primary: BKO-guided augmentation improves rare-class recall and AUC-PR by at least 10% over an unaugmented baseline, and matches or outperforms CTGAN, TVAE, and SMOTE at equal row counts.
Completeness: At least half the detected gaps close after fusion.
Validity: At least 95% of generated records pass the model's consistency checks.
Privacy: Distance-to-closest-record remains above the real-data baseline.

Because evidence on whether generative augmentation outperforms SMOTE is mixed, no-augmentation and class-weighting controls will be included, and results will be reported transparently if knowledge guidance provides no benefit.

6. Timeline and Workload Distribution

Oct 10 - Oct 22	Reproduce paper's MONDO/DO fusion; build BKO layer; train baselines.
Oct 23	Status Update #1: gap detection end-to-end; baselines locked.
Oct 24 - Nov 6	LLM repair and fusion; evaluation harness; Status Update #2.
Nov 7 - Nov 13	Streamlit UI; Working Prototype + Technical Brief.
Nov 14 - Nov 25	Final evaluation: Slides, Demo Day; Final Report.

- **Partiban: Ontology mapping, BKO inference, and gap detection.**
- **Shirley:** LLM generation, record validation, and fusion.
- **Joe:** Baseline models, evaluation metrics, and experiments.
- **Krish:** Streamlit interface, pipeline integration, and demo coordination.
- **All members:** Design reviews, results analysis, and final report/presentation.